

1M1B AI for Sustainability Virtual Internship (IBM SkillsBuild & AICTE)

MediScan AI

Scan Reports. Understand Results. Find the Right Doctor. Track Your History.



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Program : 1M1B AI for Sustainability Virtual Internship (IBM
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SDG Alignment

Primary: SDG 3 – Good Health and Well-being

- Target 3.8: Achieve universal health coverage, access to quality essential healthcare
- Target 3.4: Reduce premature mortality from non-communicable diseases via early detection

Secondary: SDG 10 (Reduced Inequalities)

- closes the healthcare access gap between urban and rural populations

SDG 9 (Industry, Innovation, Infrastructure)

- uses AI/digital infrastructure to modernize healthcare delivery



Problem Statement

"How might we use AI to help people understand their medical reports and connect with the right care so that healthcare access can become more equitable and timely?"

- Health literacy gap — most lab reports use clinical terms (e.g., "eGFR", "HbA1c") that patients can't interpret
- Referral confusion — patients often visit the wrong specialist first, wasting time and money
- Fragmented records — reports live in different apps/paper files across hospitals, no single timeline
- Delayed intervention — without someone flagging abnormal trends early, chronic conditions (diabetes, kidney disease) progress silently
- Statistic to cite (mention you'll verify/source this): India has a shortage of specialists relative to population, especially in Tier 2/3 cities and rural areas — worth a quick stat search if you want a real citation



Primary

- Patients in semi-urban/rural areas with limited access to specialists

Target Users

"I don't know which doctor to see"



Secondary

- Elderly patients managing multiple chronic conditions (diabetes, hypertension, kidney issues) who take frequent tests

Tertiary

- Caregivers/family members managing an elderly or unwell relative's care remotely

AI Solution Overview

Module 1: Report Scan & Extraction

- User uploads a photo/PDF of a lab report or scan
- OCR + document AI extracts test name, value, unit, reference range
- Handles multiple formats (pathology labs, hospitals, diagnostic chains)

Module 2: AI Summary & Explanation

- LLM converts raw values into plain-language explanation
- Flags out-of-range values with severity levels (mild/moderate/high deviation)
- Uses RAG to ground explanations in verified medical knowledge base (not free-form hallucination)

AI Solution Overview

Module 3: Doctor Matching & Booking

- Rule-based/classifier maps flagged parameters to specialist type (e.g., high TSH » Endocrinologist)
- Surfaces nearby doctors/clinics (location-based)
- Simple booking interface — select slot, confirm appointment
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Module 4: History Tracker

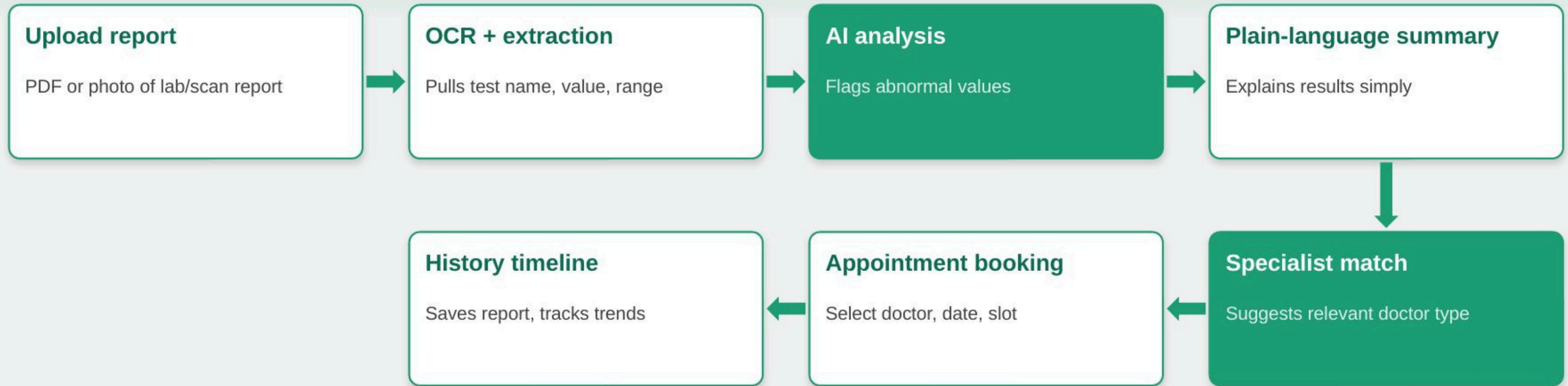
- Stores every scanned report in chronological order
- AI compares current vs. past values, visualizes trends (e.g. chart of blood sugar over 6 months)
- Alerts user to gradual changes that a single report wouldn't

Why AI Is Needed

- Information Extraction (OCR + NLP): unstructured scanned documents ➞ structured, queryable data
- Classification: maps lab parameters to relevant medical specialties using a trained/rule-based mapping table
- RAG (Retrieval-Augmented Generation): retrieves relevant medical reference content before generating explanations — reduces hallucination risk, critical in health contexts
- Pattern Detection / Time-Series Analysis: identifies trends across multiple reports that a human might miss when reviewing reports one at a time
- Why not just a static app? A rule-based lookup table alone can't handle varied report formats, phrasing, or personalized trend analysis — this is where AI adds real value over a simple database



Prototype / Workflow Diagram



Example: Creatinine 2.1 mg/dL (High, normal 0.6–1.2) → flagged → “may relate to kidney function” → Suggested specialist: Nephrologist

Responsible AI Considerations

- *Fairness: Ensure doctor/clinic database isn't urban-biased — include Tier 2/3 city and rural healthcare providers; avoid AI assuming access to specialists that don't exist locally*
- *Transparency: Every flagged result shows why (value compared to reference range), never a black-box "you have X condition"*
- *Ethics: Explicit no-diagnosis policy; AI only informs and directs, never replaces a doctor's judgment*
- *Privacy: Consent screen before any report upload*
- *Encrypted storage of medical data*
- *No sharing with third parties/advertisers*
- *Option for users to delete their history anytime*
- *Accountability: Clear disclaimer on every AI-generated summary: "This is AI-generated guidance, not a medical diagnosis."*

Expected Impact

- For patients: faster understanding of reports, reduced anxiety from medical jargon
- For healthcare access: quicker, more accurate specialist referrals reduce wasted consultations
- For continuity of care: unified history prevents repeated tests and helps doctors see the full picture in one visit
- For underserved communities: especially valuable where personal physicians or immediate specialist access isn't available
- Long-term: early trend detection (e.g., rising glucose) could support preventive care rather than reactive treatment



Thank You



"MediScan AI doesn't replace doctors — it bridges the gap between a confusing report and the right care, using AI responsibly to make healthcare a little more accessible, one report at a time."

